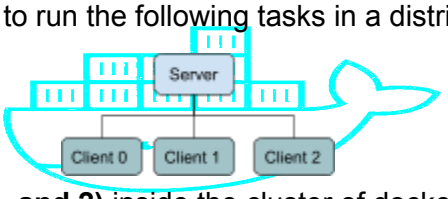
Department of Computer Science & Software Engineering

COMP-6231 Fall 2026

Assignment 3: Docker Containers And MPI

Problem Description:

Use docker containers to build a cluster of 4 docker containers in **a single host machine** (1 Master/Server and 3 worker/client) to run the following tasks in a distributed manner.



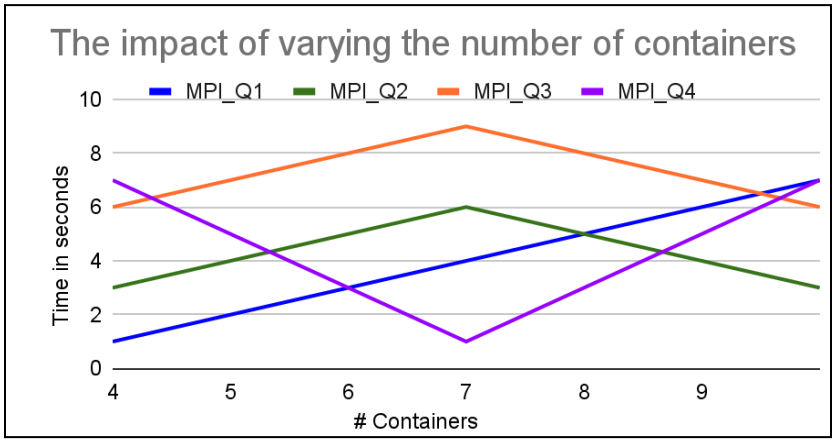
Tasks:

Run COMP6231 assignments (1, and 2) inside the cluster of docker containers.

- **Task 1 (T1):** Run Assignment 1 **file server** (or Alternative Lab Exercise in case A1 is incomplete)**.**
  - Run the file server inside the first container and the clients inside 3 other containers. Then showcase any file-tree structure for each client on the server side (.i.e., The server should create a specific directory for each client, where their respective file will be stored.)
  - List the IP/Ports used in each container and the shared storage paths if any. **(5 P)**
  - List the commands used to build the docker files and the cluster in the proper order. **(5 P)**
  - Explain each command used. **(15 P)**
- **Task 2 (T2):** Run assignment2->task3 (**parallel pandas using MPI) (or alternate MPI- Lab)**.
  - Run the master node and the 3 workers inside a cluster of 4 containers.
  - The Master node **only distributes the dataset rows across the 3 workers** and aggregates the final outputs.
  - List the IP/Ports used in each container and the shared storage paths if any. **(5 P)**
  - List the commands used to build the docker files and the cluster in the proper order and report the MPI execution results. **(20 P)**
  - Explain each used command. **(25 P)**

MPI Execution Analysis:

Prepare a report analyzing the impact of increasing the number of containers in T3 (MPI task from Assignment 2) from 4 to 10. Plot a graph with 4 lines, one for each question. The x-axis should represent the number of containers, and the y-axis should show the time taken in seconds, similar to the example figure.
(In-case the Assignment-2 is incomplete, run MPI-Pandas based exercises from the lab)



Bonus (20%): Build the cluster using **3+ physical machines** (i.e., laptops) connected via a network (i.e., virtual wifi hotspot) where the containers are distributed across the physical machines. You may use docker overl
ay network to connect the containers. (You can still earn a full grade without the bonus, your bonus will be carried to your course grade)

Docker base images (or use any other images):

- Use `docker pull ozxx33/fileserver-base` command to pull the python docker base image for task 1.
- Use `docker pull ozxx33/mpi4py-cluster-base` command to pull the mpi4py image for task 2.

Grading Distribution:

- Tasks are worth 75 points (25p for T1 and 50p for T2).
- The analysis report is worth 25 points.

Submission Instructions:

- The assignment is due at 11:59 PM on Sunday, November 1, 2026.
- Your submission should be a zip file structured as follows:

```
.zip
|___group_info.txt
|___Tasks
        |__T1_file_server
        |   |___file_server_Commands_explanation.pdf
        |__T2_parallel_pandas
            |_____MPI_Commands_explanation.pdf
            |_____MPI_analysis.pdf
```

- The Commands_explanation.pdf file should explain all commands used for the task and it must follow the same order of commands as you developed your solution. It should contain two main sections (You are not limited, but make it more readable):
    - The server/master section: lists the docker commands used to build the server/master container followed by the list of bash commands used to configure the cluster and run the task (program).
    - The client/worker section: list the docker commands used to build the client/worker container followed by the list of bash commands used to configure the cluster and run the task (program).
        - Note: Repeat this section for each client in the cluster.
    - The group_Info.txt file contains a list of group members IDs/Names. One member per line in format of : *ID<Tab> Name*
- The zip file name should have the format: <first_name>_<last_name>_<ID>_A3.zip (e.g. john_doe_11111111_A3.zip).
- If you need clarification about an unclear part of the assignment, send an email to Manush Shah (manushsunilkumar.shah@mail.concordia.ca)

- If you require help in programming, please schedule a POD session with your respective tutor and prepare your questions. The tutors may assist you with the programming and APIs but are not able to provide solutions to the assignment.
- **This is a group-based assignment consisting of (3 to 6) members. It is recommended to have the same group for this assignment as your course project**. The groups are not allowed to copy/share the solutions with other groups. Doing so is considered cheating that disqualifies the group members' submissions (0%) and may be reported to the department.
- **Each group member must submit the zip file to Moodle before the due date.**
- **Each group will have an online demo session** (to be scheduled) to showcase the working cluster (50%). Each group member will be asked 2-3 questions about their contribution to the submission (50%).

Late Policy:

- 0-24 hours late = 25% penalty.
- 24-48 hours late = 50% penalty.
- More than 48 hours late = you lose all the points for this assignment.
- Submissions of corrupted files and blank files will be considered late submissions.